

ATRIUM-9 shaft elevation -- cars A-D, floors L0 (ground) to L7 (top)

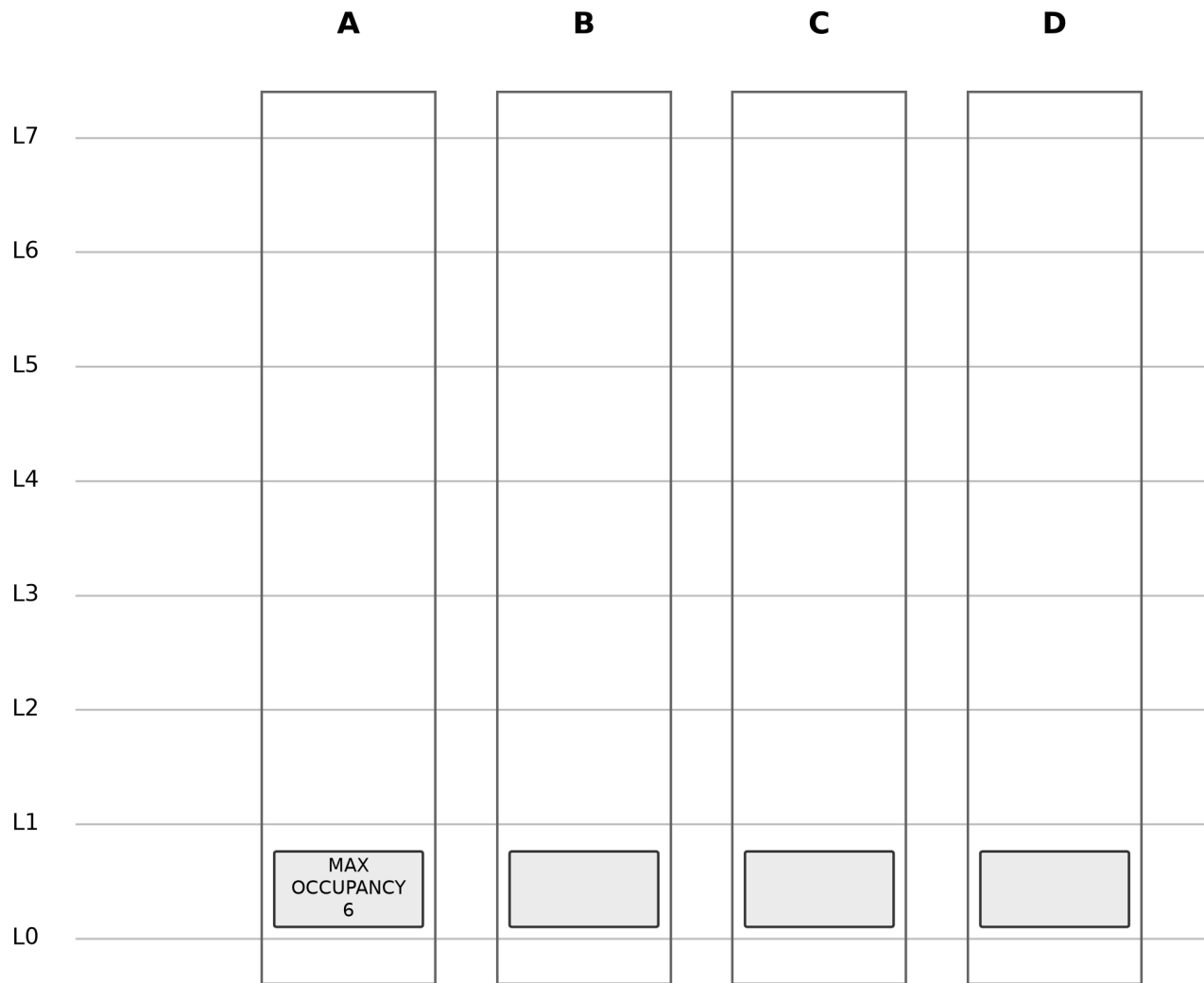


Figure 1 -- Shaft elevation. Read the occupancy placard for car capacity.

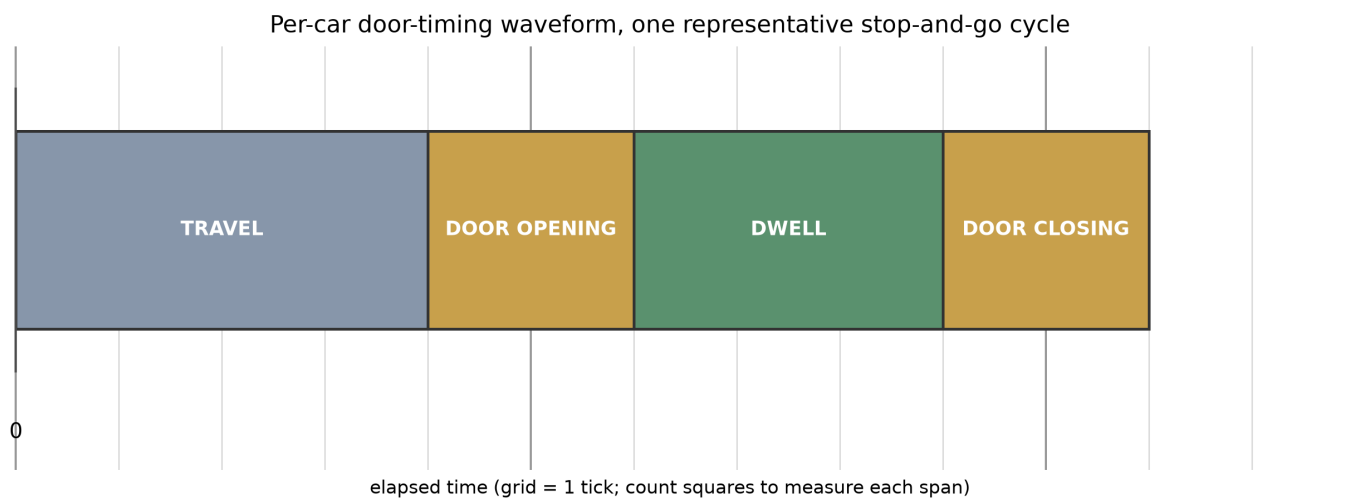


Figure 2 -- Door-timing waveform. Count grid squares to measure each phase's duration.

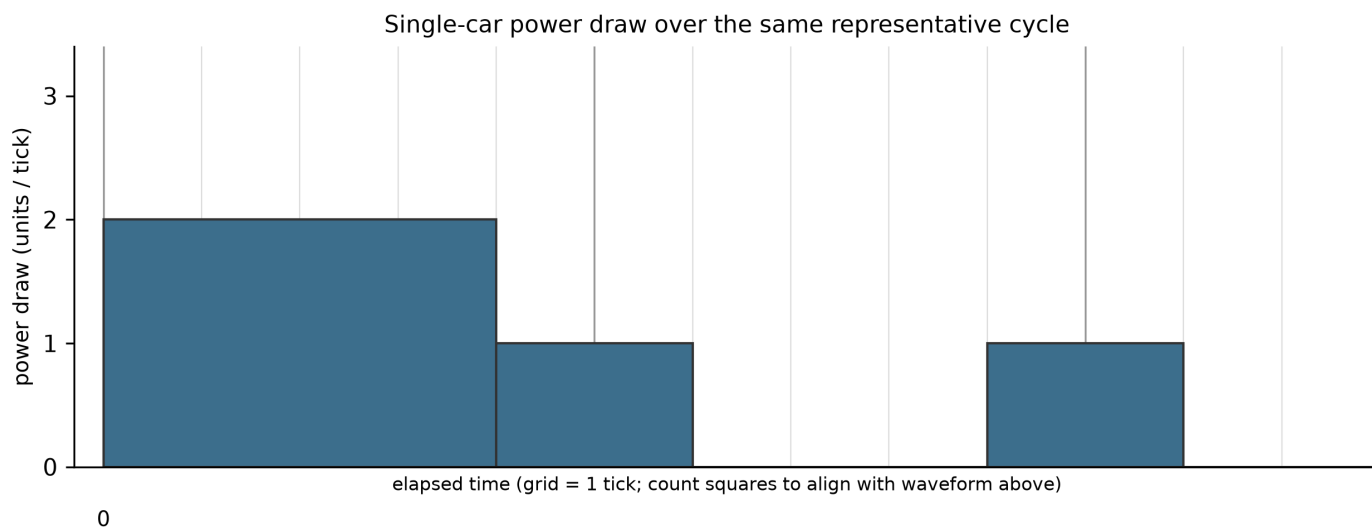


Figure 3 -- Power draw over the same cycle. Bar height = draw in units/tick.

1. WORKLOAD GENERATION

Four campaigns $s = 0..3$, twenty calls each ($k = 0..19$), 80 calls total, indexed $gid_x = 20*s + k$. Calls form bursts of three sharing an origin floor and direction: $group = k // 3$ (final group of a campaign has 2).

$origin(s,k) = (3*group + 2*s) \bmod 8$
 $direction(s,k) = UP$ if $origin == 0$
 $= DOWN$ if $origin == 7$
 $= UP$ if $(group + s)$ is even, else $DOWN$ (otherwise)
 $arrival(s,k) = 200*s + 8*k$
 $span(s,k) = 1 + ((5*k + 3*s) \bmod 7)$
 $dest(s,k) = \min(7, origin+span)$, or $\min(7, origin+1)$ if that
 equals origin, when direction is UP
 $= \max(0, origin-span)$, or $\max(0, origin-1)$ if that
 equals origin, when direction is $DOWN$

A call's destination is revealed only at boarding, not at hall-call assignment time.

2. DISPATCH POLICY (fully deterministic -- no free per-tick choice)

A car is eligible for a pending hall call (floor, direction) iff it is idle (cost = distance in ticks), or it is currently scanning that same direction AND the floor still lies ahead of its position in that direction. A car scanning the opposite direction, or one that has already passed the floor, is not eligible this tick -- no backtracking mid-scan. Assign to the minimum-cost eligible car; ties break to the lowest car ID. Each car tracks its committed stops SEPARATELY per scan direction -- a floor needed while scanning UP is a different commitment from the same floor needed while scanning $DOWN$. A car re-derives its next action fresh every tick: continue the current scan direction while a commitment remains ahead in it, else switch to the nearer remaining commitment, else head to its home floor (zoning table below) and idle.

3. DOORS AND POWER (see the waveform and power-draw diagrams)

One full stop is: $TRAVEL$ (per floor traversed) $\rightarrow$ $DOOR\ OPENING$ $\rightarrow$ $DWELL$ $\rightarrow$ $DOOR\ CLOSING$ $\rightarrow$ $CLOSED$. Boarding and alighting happen atomically at the first $DWELL$ tick, ascending call-index order, up to the fixed car capacity shown on the shaft elevation. $DWELL$ itself draws no power. Building power is a $SHARED$ cross-car budget (a sweep knob): a car mid-travel or mid-door-cycle keeps the power it was already admitted; starting a NEW travel hop, starting door $OPENING$, or the transition from $DWELL$ to $CLOSING$ once dwell has elapsed are each their own power-gated event, admitted each tick from whatever budget remains, in fixed car priority A, B, C, D. The $DWELL$ -to- $CLOSING$ transition is NOT automatic: a car that has finished dwelling stays in $DWELL$, drawing no power, until its door-close is itself admitted that tick.

4. ZONING (idle parking policy -- a sweep knob)

SPLIT: A->L0 B->L2 C->L5 D->L7

GROUND: A->L0 B->L0 C->L0 D->L0

TOP: A->L7 B->L7 C->L7 D->L7

Only the lowest-ID ACTIVE_CARS cars are in service (a sweep knob, 2, 3, or 4); an inactive car never moves and is never eligible.

5. TIMEOUT REASSIGNMENT (non-combinable, cost-compared)

Every tick, for any still-unboarded call whose current assignment has aged past WAIT_TIMEOUT ticks (a sweep knob: 50, 75, 100, or 125), find the best alternative eligible car excluding the incumbent. Reassign only if that alternative is strictly cheaper than the incumbent's own recomputed cost -- never bounce a call between equally good cars. A call reassigned away from a car is never later served by that same car.

6. ENERGY

Every tick a car is mid-travel: energy += 3 + load if moving up, energy += 3 - load if moving down (regenerative credit for a loaded descent). net_energy is the running total over the whole run.

7. THE SEARCH -- 144-configuration sweep

The legal configuration space is ACTIVE_CARS x ZONING x WAIT_TIMEOUT x POWER_BUDGET = {2,3,4} x {SPLIT,GROUND, TOP} x {50,75,100,125} x {6,8,10,12} = 144 configurations; capacity is fixed (see the shaft elevation) and not swept. Run every configuration as one continuous 80-call simulation. Report three lexicographic selections:

- wait-optimal: minimize avg_wait, tie-break net_energy
- energy-optimal: minimize net_energy, tie-break avg_wait
- budget-constrained: minimize avg_wait among configurations with net_energy <= 3800, tie-break net_energy

These three selections are not required to agree.

8. CERTIFICATION

For the baseline configuration (active_cars=3, zoning=TOP, wait_timeout=50, power_budget=8) only: serialize the complete per-tick trace as one line per tick "t,power,energy" in execution order (energy is that tick's own signed contribution, not a running total), preceded by one header line "CONFIG:{'active_cars': 3, 'zoning': 'TOP', 'wait_timeout': 50, 'capacity': <see shaft elevation>, 'power_budget': 8}" using Python dict.__str__ formatting and this exact key order, lines joined by \n, UTF-8 encoded, hashed with SHA-256; report the first 16 hex characters as the trace-integrity certificate.

Deliver: simulator source, an independently-coded verifier, the full 144-row sweep table, a decision file with all three selections, the baseline trace plus its SHA-256 certificate and your adversarial-mutation results, and an engineering memo. Base every reported number on your own executed implementation.